

# Tractor

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# Author's note

You should watch the [YouTube video](#) where I present these slides.

- Trick-taking card game popular in Chinese communities
- Goes by many names: *tuo la ji* ( 拖拉机, literally “tractor”), *sheng ji* ( 升级, literally “upgrade”), *ba shi fen* ( 八十分, literally “eighty points”)

# This presentation

- Includes lots of examples, but also concise
- Sufficient by itself
- Originally produced for RSI 2026
- Teaching is hard

# Conventions for experts

- Two decks, four players.
- No throw penalty.
- Kitty points multiplied by  $2^n$ , where  $n$  is the size of the largest component of the last trick.
- Subsequent throw must beat all cards to win.
- A must be defended, but 5, 10, and K do not need to be.
- J variation. Hehe.

# High-level summary

- 2v2 card game; four players and two decks (incl. low joker LJ, high joker HJ)
- Teammates sit across from each other
- Teams start at rank 2; they advance through the ranks 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A, NT (no trump) in order
- Winning rounds moves you up ranks; how many ranks you go up depends on “how much you win”
- Whoever passes rank NT first wins the full game

# High-level summary

- In each of ♣, ♦, ♥, ♠: 5, 10, K worth 5, 10, 10 points, respectively
- Everyone gets 25 cards
- A *trick*: the “lead” plays something, the players proceed counterclockwise **matching format**. Whichever team plays the highest thing of that format wins all points played in that trick.
- Tricks proceed until everyone is out of cards
- Goal: maximize points scored by your team

# Example

## Convention

Examples have **Obama** and **Biden** (the *Democrats*) play **Trump** and **Reagan** (the *Republicans*).

Here, **Obama** is lead.

- **Obama** plays A♣.
- **Trump** plays 4♣.
- **Biden** plays 10♣.
- **Reagan** plays A♣.

**Obama** wins the trick since his A♣ takes priority over **Reagan**'s. The Democrats win 10 points from the 10♣.

The *host* or *landlord* has two advantages.

- They make the first play.
- After everyone draws 25 cards, the host picks up the 8 remaining cards (the “kitty”) and replaces it with their 8 worst ones.

The host's team is called the *defenders*; the other team is called the *attackers*. Who's host?

- Next person counterclockwise on the winning team is new host
- Host for first round is decided through bidding

# The trump suit

The *trump suit* is the **highest suit**. Suits that are not the trump suit are called **offsuits**.

Plays matching shape in the trump suit beat corresponding plays in offsuits (“trumping”).

The trump suit consists of:

- Every joker
- Every card in the trump rank
- Every card in either ♣, ♦, ♥, ♠, depending on bidding

# Example

When the trump rank is 4 and the trump suit is ♠ (colloquially, when the trump suit is 4♠), the trump suit is

2♠3♠5♠6♠7♠8♠9♠10♠J♠Q♠K♠A♠4♣4♦4♥4♠LJHJ

and the heart suit is

2♥3♥5♥6♥7♥8♥9♥10♥J♥Q♥K♥A♥.

Above, the cards are in strictly increasing order in their suits, except 4♣, 4♦, and 4♥ are equal. So generally, 2 is the lowest and A is the highest.

# A reminder

When the trump suit is 4♠, the trump suit is

2♠3♠5♠6♠7♠8♠9♠10♠J♠Q♠K♠A♠4♣4♦4♥4♠LJHJ

and the heart suit is

2♥3♥5♥6♥7♥8♥9♥10♥J♥Q♥K♥A♥.

**WHEN THE TRUMP SUIT IS 4♠, 4♥ IS NOT IN THE HEART SUIT!!!**

It is in the trump suit consisting of spades, 4s, and jokers.

Slogan: “4♥ is a ♠.”

# Example

4♠ is trump.

- Obama plays 4♦.
- Trump plays 4♣.
- Biden plays 10♠.
- Reagan plays 5♠.

The Democrats win the trick and 15 points.

# Example

4♠ is trump.

- Obama plays 4♦.
- Trump plays 4♠.
- Biden plays 10♠.
- Reagan plays 5♠.

The Republicans win the trick and 15 points.

# Types of plays

*the “lead” plays something...*

What is “something?”

- A single
- A pair
  - Pairs:  $9\spadesuit 9\spadesuit$ . HJHJ.
  - Not pairs: when  $6\spadesuit$  trump,  $2\heartsuit 2\heartsuit$ ,  $6\heartsuit 6\heartsuit$ , LJHJ.
- A tractor (sequences of consecutive pairs)
  - Tractors: when  $6\spadesuit$  trump,  $9\spadesuit 9\spadesuit 10\spadesuit 10\spadesuit$ ,  $5\heartsuit 5\heartsuit 7\heartsuit 7\heartsuit$ ,  $A\spadesuit A\spadesuit 6\heartsuit 6\heartsuit 6\spadesuit 6\spadesuit$  LJLJHJHJ
  - Not tractors: when  $6\spadesuit$  trump,  $A\heartsuit A\heartsuit 2\heartsuit 2\heartsuit$ ,  $6\heartsuit 6\heartsuit 6\spadesuit 6\spadesuit$

In particular, *you cannot play straights.*

Bidding occurs while drawing. Two purposes:

- Decide host for first round
- Decide the *trump suit*

In increasing order, the possible bids are a single of the trump rank, a pair of the trump rank, a pair LJ, and a pair HJ. You keep the cards you bid.

You can't overturn your own bid, but you can reinforce a bid of a single by making it into a pair.

# No trump

If a single/pair of the trump rank is bid, the suit of the winning bid is just the trump suit. But if a pair of jokers is bid, the trump suit is *no trump* (NT). This means there are *five* suits: ♣, ♦, ♥, ♠, and the trump suit consisting of the cards of the trump rank and jokers.

# Example

When the trump suit is 4NT, the trump suit is

$4\clubsuit 4\diamondsuit 4\heartsuit 4\spadesuit \text{LJHJ}$

where  $4\clubsuit = 4\diamondsuit = 4\heartsuit = 4\spadesuit < \text{LJ} < \text{HJ}$ , and the club suit is

$2\clubsuit 3\clubsuit 5\clubsuit 6\clubsuit 7\clubsuit 8\clubsuit 9\clubsuit 10\clubsuit \text{J}\clubsuit \text{Q}\clubsuit \text{K}\clubsuit \text{A}\clubsuit$

in increasing order.

# Example

Trump rank 2, first round.

- Obama bids 2♣.
- Reagan overturns with 2♠2♠.

There are no more overturns. Reagan is host with trump suit 2♠.

# Example

Trump rank 4.

- Obama bids 4♣.
- Obama reinforces with 4♣4♣.
- Trump bids LJLJ.
- Obama bids HJHJ.

The trump suit is 4NT.

# Matching format

*the players proceed counterclockwise **matching format**...*

What is “matching format?”

- First priority: match suit
- Second priority: match shape

**If you can't match suit, you don't need to match shape.** Players always play the same number of cards.

# Example

$10\spadesuit$  is trump.

- Obama plays  $A\clubsuit$ .
- Trump's  $\clubsuit$ :  $5\clubsuit 5\clubsuit K\clubsuit$ . He plays  $5\clubsuit$ .
- Biden has no  $\clubsuit$ . He trumps it with  $K\spadesuit$ .
- Reagan has no  $\clubsuit$ . He trumps it with  $10\clubsuit$ .

Reagan's  $10\clubsuit$  wins 25 points for the Republicans.

# Example

10♦ is trump.

- Obama plays K♥.
- Trump plays 2♥.
- Biden has no ♥ and plays A♠.
- Reagan plays 5♥.

Obama wins the trick and 15 points.

# Example

10♦ is trump.

- Obama plays 4♦.
- Trump plays 4♦.
- Biden has no ♦, 10s, or jokers, and plays 7♥.
- Reagan plays 2♦.

Obama wins the trick and 0 points.

# Example

8♦ is trump.

- Obama plays K♥K♥.
- Trump's ♥: 2♥3♥5♥5♥9♥. He plays 5♥5♥.
- Biden's ♥: 3♥7♥7♥10♥J♥. He plays 7♥7♥.
- Reagan's ♥: 6♥A♥. He plays 6♥A♥.

Obama wins the trick and 30 points.

# Example

8♦ is trump.

- Obama plays K♥K♥.
- Trump has no ♥ and trumps it with 10♦10♦.
- Biden has exactly one ♥ left: the 10♥. He plays 10♥8♠.
- Reagan is out of ♥ and plays 10♣10♠.

Trump wins the trick and 70 points.

# Example

8♦ is trump.

- Obama plays K♥K♥.
- Trump has no ♥ and tries to trump it with 6♦7♦.
- Biden's ♥: 10♥. He plays 10♥10♠.
- Reagan plays 2♣4♣.

Obama wins the trick and 40 points.

# Example

8♦ is trump.

- Obama plays 7♥7♥9♥9♥.
- Trump's ♥: 6♥10♥K♥. He plays 6♥10♥K♥4♠.
- Biden's ♥: 2♥3♥3♥10♥K♥A♥A♥. He plays 3♥3♥A♥A♥.
- Reagan has no ♥ and plays A♦A♦8♥8♥.

Reagan wins the trick and 20 points.

# Example

8♦ is trump.

- Obama plays 2♠2♠.
- Trump plays 4♠Q♠.
- Biden plays 6♠A♠.
- Reagan plays 3♠3♠.

Reagan wins the trick and 0 points.

# Scoring rules

| Attacker points  | Name           | Result       |
|------------------|----------------|--------------|
| 0                | sweep          | defenders +3 |
| [5, 35]          |                | defenders +2 |
| [40, 75]         |                | defenders +1 |
| [80, 115]        | stalemate      | no change    |
| [120, 155]       |                | attackers +1 |
| [160, 195]       |                | attackers +2 |
| [200, $\infty$ ) | attacker sweep | attackers +3 |

Stalemate and above is considered a win for the attackers, in which case the attacking team becomes the new host.

# Throwing

*Throwing* allows you to play multiple things at the same time, as long as nothing played can be beaten within the suit.

After a throw is played, players affirm in counterclockwise order they can't defeat any component. If the throw's invalid, the component that can be defeated is played. If multiple components can be defeated, the first player (in counterclockwise order) which can defeat any component chooses a component they can beat; that component is played.

In order to trump or re-trump a throw, you need to strictly beat every component.

# Example

4♥ is trump.

- Obama throws A♠Q♠Q♠. Everyone affirms this throw is valid.
- Trump plays 10♠2♥2♥.
- Biden plays 5♠6♠6♠.
- Reagan plays 3♠6♦7♦.

Obama wins the trick and 15 points.

# Example

4♥ is trump.

- **Obama** throws A♠9♠9♠10♠10♠. **Trump** says the throw is invalid, so only the 9♠9♠10♠10♠ goes through. **Obama** takes back the A♠.
- **Trump** plays Q♠Q♠K♠K♠.
- **Biden** plays 2♠2♠5♠8♦.
- **Reagan** plays 5♠6♠6♠7♠.

**Trump** wins the trick and 50 points.

# Example

4♥ is trump.

- **Obama** throws A♠K♠. Assume an A♠ was already played, so this throw is valid.
- **Trump** plays 10♥K♥.
- **Biden** plays K♥HJ.
- **Reagan** plays 6♦7♦.

**Biden** wins the trick and 40 points.

# Kitty multiplier

*After everyone draws 25 cards, the host picks up the 8 remaining cards (the “kitty”) and replaces it with their 8 worst ones.*

Whoever wins the last trick wins all the points in the kitty. Additionally, the points are multiplied by  $2^n$ , where  $n$  is the size of the largest component of the last trick.

- With a single, kitty points are  $\times 2$ 'd.
- With a pair, kitty points are  $\times 4$ 'd.
- With a tractor, kitty points are  $\times 16$ 'd.

# Finding friends

- *Zhao peng you* (找朋友, literally “finding friends”), also called “zpy” or “ff”: a variant allowing the game to be played with an odd number of players
- With  $n$  players, landlord has  $f := \lfloor \frac{n}{2} \rfloor - 1$  friends
- Make  $f$  calls of the form “ $i$ th card  $c$  played is my teammate” ( $c$  can’t be trump)
  - Examples: first A♣ or second A♠
- Host and friends can join team multiple times
- Small team bonus: if this happens and host wins, their team goes up an extra rank
- Despite this, small teams suck
- 7p7d zpy is the greatest form of Tractor

# Example

2NT is trump. 5p5d zpy. Host called 3rd A♠. First play.

- Obama throws  $3\spadesuit 3\spadesuit 4\spadesuit 4\spadesuit 5\spadesuit 5\spadesuit 6\spadesuit 6\spadesuit 7\spadesuit 7\spadesuit 8\spadesuit 8\spadesuit A\spadesuit$ . Biden says the throw fails.
- Trump plays  $3\spadesuit 4\spadesuit 4\spadesuit 5\spadesuit 7\spadesuit 8\spadesuit 9\spadesuit 9\spadesuit 10\spadesuit J\spadesuit Q\spadesuit A\spadesuit$ .
- Biden plays  $8\spadesuit 8\spadesuit 9\spadesuit 9\spadesuit 10\spadesuit 10\spadesuit J\spadesuit J\spadesuit Q\spadesuit Q\spadesuit K\spadesuit K\spadesuit$ .
- Reagan plays some 12 cards.
- Bush plays some 12 cards.

Biden wins the trick.

# SKISSY SQRT!

The above *actually happened*. The host was forced to self-friend and got swept.

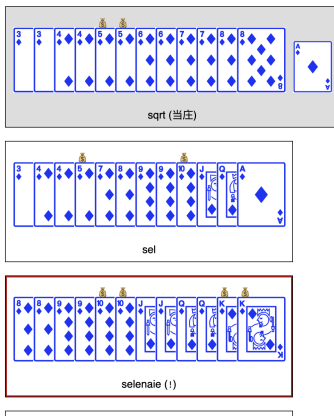


Figure: SKISSY SQRT!

# Single J rule

If the defenders are rank J and the attackers win the last trick with a single J, the defenders go back to rank 2. Any rank change from that round still occurs after the defenders go back to rank 2.

# Basic strategy

- Play offsuit aces, high pairs, then low pairs. If you have nothing, play low trump to try and transfer control to your teammate.
- Try to get rid of offsuits.
- Play suits your teammate is out of. Don't play suits your opponents are out of.
- Try not to break up pairs.
- Don't throw only singles in suits your opponents are out of.
- If opponents play points/aces in offsuits, they're likely almost out.

- On <https://robertying.com/shengji/>, parts of this presentation are adapted from <https://robertying.com/shengji/rules.html>
- Forces you to follow the rules
- Has “help” button telling you what you’re allowed to play
- Has “BEEP” button to tell others to hurry up
- Allows for 1v1s via splitscreen

# An example game

<https://robertying.com/shengji/>